

## Raw SERS data



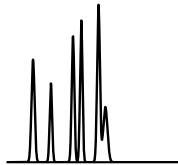
**SERS map**

one spectrum  
per pixel



**NNLS gate**

background  
dropped



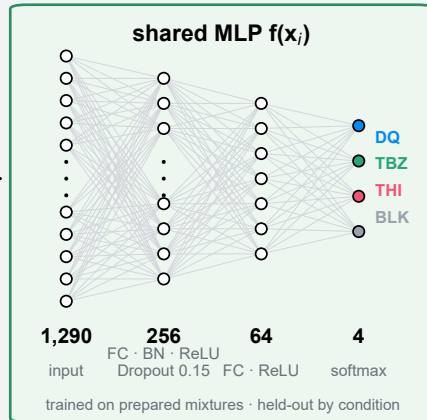
**pixel spectra  $x_i$**

1,290 pts · 500–2500  $\text{cm}^{-1}$

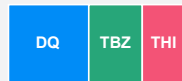


## Compositional unmixing

output



### map composition



gated pixels · composition

$\bar{p}_i = \text{pixel mean} \cdot \text{BLK removed}$

absent component → reported 0

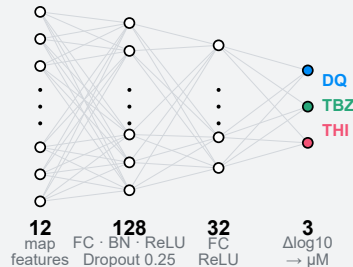
(absence gate,  $P < 0.2$ )

## Concentration

output

12 features:  $\log_{10} C_{\text{cal}} \cdot 3 \cdot \text{composition} \cdot 3 \cdot \log_{10} \text{band} \cdot 3 \cdot \log_{\text{total}} \text{p10/50/90}$

### concentration net $g(\text{map})$



$C = C_{\text{cal}} \cdot 10^{\Delta}$ ;  $C_{\text{cal}}$  = log-linear band calibration (9–144  $\mu\text{M}$ )  
reported only if the nearest training map is within z-distance 3 · else refused

**declared total (optional)**

$\hat{C}_i = \bar{p}_i \cdot C_{\text{total}}$  — overlay when known